



CFG TO CNF

CFG

CFG stands for context-free grammar. It is a formal grammar which is used to generate all possible patterns of strings in a given formal language.

CFG

Context-free grammar G can be defined by four tuples as

$$G = (V, T, P, S)$$

CFG

G is the grammar, which consists of a set of the production rule. It is used to generate the string of a language.

T is the final set of a terminal symbol. It is denoted by lower case letters.

V is the final set of a non-terminal symbol. It is denoted by capital letters.

P is a set of production rules, which is used for replacing non-terminals symbols(on the left side of the production) in a string with other terminal or non-terminal symbols(on the right side of the production).

S is the start symbol which is used to derive the string. We can derive the string by repeatedly replacing a non-terminal by the right-hand side of the production until all non-terminal have been replaced by terminal symbols.

CNF

CNF stands for Chomsky normal form. A CFG(context free grammar) is in CNF(Chomsky normal form) if all production rules satisfy one of the following conditions:

- Start symbol generating ϵ . For example, $A \rightarrow \epsilon$.
- A non-terminal generating two non-terminals. For example, $S \rightarrow AB$.
- A non-terminal generating a terminal. For example, $S \rightarrow a$.

CONVERTTING CFG TO CNF



Steps for converting CFG into CNF

- Step 1: Eliminate start symbol from the RHS. If the start symbol T is at the right-hand side of any production, create a new production as: $S_1 \rightarrow T$
- Where S_1 is the new start symbol.
- Step 2: In the grammar, remove the null, unit and useless productions. You can refer to the Simplification of CFG.
- Step 3: Eliminate terminals from the RHS of the production if they exist with other non-terminals or terminals. For example, production $S \rightarrow aA$ can be decomposed as:
 - $S \rightarrow RA$
 - $R \rightarrow a$
- Step 4: Eliminate RHS with more than two non-terminals. For example, $S \rightarrow ASB$ can be decomposed as:
 - $S \rightarrow RS$
 - $R \rightarrow AS$

EXAMPLE

Convert the given CFG to CNF. Consider the given grammar G1:

$$S \rightarrow a \mid aA \mid B$$
$$A \rightarrow aBB \mid \varepsilon$$
$$B \rightarrow Aa \mid b$$

SOLUTION:

Step 1: We will create a new production $S1 \rightarrow S$, as the start symbol S appears on the RHS. The grammar will be:

$$S1 \rightarrow S$$

$$S \rightarrow a \mid aA \mid B$$

$$A \rightarrow aBB \mid \varepsilon$$

$$B \rightarrow Aa \mid b$$

Step 2: As grammar G1 contains $A \rightarrow \epsilon$ null production, its removal from the grammar yields:

$S1 \rightarrow S$

$S \rightarrow a \mid aA \mid B$

$A \rightarrow aBB$

$B \rightarrow Aa \mid b \mid a$

Now, as grammar G1 contains Unit production $S \rightarrow B$, its removal yield:

$S1 \rightarrow S$

$S \rightarrow a \mid aA \mid Aa \mid b$

$A \rightarrow aBB$

$B \rightarrow Aa \mid b \mid a$

Also remove the unit production $S1 \rightarrow S$, its removal from the grammar yields:

$S0 \rightarrow a \mid aA \mid Aa \mid b$

$S \rightarrow a \mid aA \mid Aa \mid b$

$A \rightarrow aBB$

$B \rightarrow Aa \mid b \mid a$

Step 3: In the production rule $S0 \rightarrow aA \mid Aa$, $S \rightarrow aA \mid Aa$, $A \rightarrow aBB$ and $B \rightarrow Aa$, terminal a exists on RHS with non-terminals. So we will replace terminal a with X :

$$S0 \rightarrow a \mid XA \mid AX \mid b$$
$$S \rightarrow a \mid XA \mid AX \mid b$$
$$A \rightarrow XBB$$
$$B \rightarrow AX \mid b \mid a$$
$$X \rightarrow a$$

Step 4: In the production rule $A \rightarrow XBB$, RHS has more than two symbols, removing it from grammar yield:

$$S0 \rightarrow a \mid XA \mid AX \mid b$$
$$S \rightarrow a \mid XA \mid AX \mid b$$
$$A \rightarrow RB$$
$$B \rightarrow AX \mid b \mid a$$
$$X \rightarrow a$$
$$R \rightarrow XB$$

Hence, for the given grammar, this is the required CNF.

END
OF
Presentation

**THANK
YOU**